JOINT ENTRANCE EXAMINATION (JEE MAIN

Paper / Shift ID: Session 1 Shift 2
Duration: 3 Hours
Maximum Marks: 300

Instructions to Candidates:

1. This paper contains sections for Physics, Chemistry, and Mathematics.

2. Candidates must follow the instructions given per question type.

3. MCQ questions have positive and negative markings as indicated.

4. Numerical and Integer value questions do not have negative marking.

Section 1: Physics

Q1. [MCQ-single] [+4, -1]

A block of mass $m = 2$ kg is placed on a rough inclined plane making an angle of 30 degrees with the horizontal. If the coefficient of static friction is 0.5, what is the friction force acting on the block?

(A) 9.8 N
(B) 4.9 N
(C) 8.5 N
(D) 19.6 N

Q2. [Numerical] [+4, 0]

A parallel plate capacitor has plate area A and separation d. A dielectric slab of constant $K = 4$ is inserted to fill half the volume. Find the equivalent capacitance in terms of C_0.

Section 2: Chemistry

Q1. [MCQ-single] [+4, -1]

Which of the following organic molecules exhibits geometric isomerism due to restricted rotation about a carbon-carbon double bond?

(A) But-2-ene
(B) Propene
(C) But-1-ene
(D) 2-Methylpropene

Q2. [Numerical] [+4, 0]

For a first order reaction, the half-life is 1386 seconds. Calculate the rate constant of the reaction in s^{-1} multiplied by 10^4.

Section 3: Mathematics

Q1. [MCQ-single] [+4, -1]

Let the function $f(x) = x^3 - 3x + 2$. Find the total number of local extrema of $f(x)$ on the real line.

(A) 0
(B) 1
(C) 2
(D) 3

Q2. [Numerical] [+4, 0]

Evaluate the definite integral of $x * \exp(x)$ from 0 to 1. Round to the nearest integer.